Que 7) Given two languages L_1 and L_2 defined over $\Sigma = \{a, b\}$, L_1 accepts palindrome strings and L_2 accepts string with equal number of 0's and 1's which one of these language is regular? Give reasons.

⇒ (1) A "palindrome" reads the same forward & backward

e.g.:- ϵ, a, aa, aba, abba

→ To recognize a palindrome, a machine must compare the first symbol with the last, second with second-last etc.

→ A finite Automata (FA) can't remember unlimited symbol to compare later.

∴ L_1 is not regular language

(2) L_2 : equal no of 0's & 1's

e.g.:- 01, 10, 0011, 0101, ...

→ Here the machine must count the no of 0's & 1's & ensure they are equal.

→ A finite Automata (FA) has no memory to count unlimited symbols.

∴ L_2 is not Regular language

Que) Construct a finite Automata that accepts all strings containing 010 or 111 as sub-string only